

A-060

Roll No.

038

Total No. of Pages.: 03

21N508

B.Tech. II Sem Main/Back (New Scheme) Acad. Session 2023-24

All Branch

**(2FY2-08) – Computer Fundamentals & Programming
Common to All Branches**

21N508

Time : 3 Hours

Maximum Marks: 70

Min. Passing Marks:

Instructions to Candidates:

Part – A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All 10 questions are compulsory.

Part – B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving/ Design questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

P.T.O.

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PART-A

- Q.1. Draw a flowchart to check whether a number is even or odd?
- Q.2. What is output conversion of $(1010.267)_8 = (?)_2$
- Q.3. What do you mean by else if ladder? Explain with suitable example?
- Q.4. Explain inter function communication?
- Q.5. What is the difference between text file and binary file?
- Q.6. Evaluate the expression $a = 7/22 * (3.14+2) * 3/5$.
- Q.7. What are literals? Explain in brief?
- Q.8. What do you mean by primary & secondary memory?
- Q.9. Explain the concept of array in C?
- Q.10. Define recursion?

PART-B

- Q.1. What do you understand by precedence of operator? How it is considered in expression evaluation?
- Q.2. Explain representing algorithm through a flow chart?
- Q.3. Write a C program to reverse an integer inputted number e.g. if input is 2345 then output should be 5432.
- Q.4. What is void pointer? Discuss its uses?
- Q.5. Write a program to print following pattern?
- ```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```
- Q.6. List out various iterative statements used in C?
- Q.7. Differentiate between compiler and Interpreter?

### PART- C

- Q.1. What is a pseudocode? How does it differ from flowchart? Draw and explain various symbols of flowchart for finding a greatest among 3 numbers?
- Q.2. Explain the basic architecture of computer with a neat diagram.
- Q.3. What is a structure? Define a structure called book with book name, Author name and price. Write a C program to read the details of book name, author name and price of 200 books in a library and display the total cost of books?
- Q.4. Convert the following numbers from one system to another:
- a.  $(123.63)_{10} = (?)_2$
  - b.  $(AB20C.00C)_{16} = (?)_{10}$
  - c.  $(651.24)_8 = (?)_2$
  - d.  $(7896)_{10} = (?)_{16}$
  - e.  $(5272)_{10} = (?)_8$
- Q.5. Explain the concept of 1-D and 2-D array? Write a C program to perform matrix addition operation?

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